Wall time per step: Parse 1M PBMC, 500k (346,259 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	55	27	30	23	OOM	12	23	OOM	162	366	OOM	OOM	70
qc	1.9	1.9	1.4	1.3	–	3.8	4.0	–	2.2	5.6	–	–	4.2
normalise	n/a	n/a	n/a	n/a	–	3.5	0.0	–	0.4	0.0	–	–	2.5
hvg	0.9	1.1	0.8	0.8	–	3.2	5.0	–	0.9	13	–	–	14
pca	2.5	2.4	0.9	1.3	–	5.7	9.8	–	10	22	–	–	350
harmony	2.1	2.2	1.3	0.6	–	2.4	2.3	–	87	124	–	–	92
knn_snn	11	11	5.8	5.2	–	31	27	–	26	28	–	–	33
cluster	4.0	4.1	3.3	3.1	–	14	13	–	89	77	–	–	111
cluster_fast	12	12	1.9	1.6	n/a	n/a	n/a	–	117	118	n/a	n/a	n/a
umap	22	22	2.5	2.4	–	155	153	–	150	152	–	–	198
dge	2.6	2.6	2.4	2.3	–	602	–	–	36	1,396	–	–	953
tsne_default	150	149	132	129	–	796	806	n/a	n/a	n/a	–	–	598
tsne_fft	37	38	9.5	13	–	109	109	n/a	n/a	n/a	–	–	110
tsne_quick	24	25	14	13	n/a	n/a	n/a	–	96	101	n/a	n/a	n/a
total (no tSNE)	102	75	48	40	failed	832	237	failed	565	2,182	failed	failed	1,829

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.